

ECOG 315 / ECON 181, Summer 2025
Advanced Research Methods and Statistical Programming
Week 7 Lecture Slides

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Today's Agenda

1. Structural modeling: what's so difficult about that?
2. Comments on project management and research collaboration
3. Advice for your final presentations
4. Questions / discussion / feedback on student projects

What Makes MicroDSOPs “Difficult” ?

- ▶ Additional state and control dimensions: **curse of dimensionality**
- ▶ FOCs necessary but not sufficient to characterize solution
- ▶ Mixture of discrete and continuous control variables
- ▶ “Bad boundary behavior”: infinities at lower bound, difficult extrapolation
- ▶ *Ex ante* heterogeneity: agents don’t all have same parameters
- ▶ Debugging: complex objects, hard to find where bug starts; need to know theory
- ▶ Approximation: how close is close enough?
- ▶ Simulation: necessary Monte Carlo population can be large
- ▶ “Macro” layer: things determined in equilibrium from micro model

What Goes into a Structural Economics Project? (1/2)

1. Identify a topic area and question(s) of interest
2. Justify why this calls for a structural model (rather than reduced form)
3. Design the model, maybe by adapting / expanding prior model
4. Characterize model solution on paper, understand theory & mechanisms
5. Program the microeconomic solver: parameters $\rightarrow$ solution
6. Program the microeconomic simulation: solution $\rightarrow$ outcomes
7. Determine which parameters can be set or calibrated without using the model
8. Gather & process data for reduced form sub-estimations for some of those
9. Devise a **structural identification strategy** for remaining parameters

What Goes into a Structural Economics Project? (2/2)

10. Gather and process data needed for structural estimation
11. Program the objective function (MLE, SMM, etc): outcomes $\longrightarrow$ model fit
12. Estimate structural parameters by optimizing objective function
13. Compute standard errors by appropriate method; is model well identified?
14. Evaluate (and demonstrate) fitness of model to data; does it actually work?
15. Devise interesting counterfactual experiments and other analysis to use the model
16. Program and execute those experiments and analyses
17. Interpret and explain those results: how and why did the model behave like that?
18. Figure out how to coherently explain all of this in a 25-90 min presentation

So You're Going to Coauthor a Paper (1/2)

- ▶ Determine whether this needs to be coauthored; why can't it be done solo?
- ▶ Make a concrete research plan and outline for the entire project; include dates
- ▶ Decide who will do what, and when they will do it; do they have time for that?
- ▶ Begin project organization **immediately**, don't procrastinate on it
- ▶ All files should be in a single place (with subdirectories) and **backed up**
- ▶ Have (semi-)regularly scheduled meetings on project progress; stay accountable
- ▶ Maybe: open a GitHub issue for your current project task(s) for visibility
- ▶ Be descriptive in git commit names and comments; not "edited document"
- ▶ Put comments on **everything**; I promise you will not remember later

So You're Going to Coauthor a Paper (2/2)

- ▶ Be explicit and clear in comments; don't be lazy. Put a docstring on everything
- ▶ Discuss what journal(s) you think this paper might realistically get into
- ▶ Communicate about problems as they come up, don't let resentments build
- ▶ Establish criteria / procedures early about whether / how to add new coauthor
- ▶ Live coding the same file? Unnecessary; see “two idiots one keyboard”
- ▶ Live **text** collaboration on HackMD, Quip, or Google Docs is fine
- ▶ Code / automate almost everything, e.g. reduced form results as model input
- ▶ For big projects: use automated tested / continuous integration
- ▶ For big projects: clear expectations and procedures for handling PRs

Giving Your Research Presentation (1/2)

- ▶ The audience doesn't necessarily remember anything; start fresh
- ▶ Your time budget is **even tighter** than before; keep it moving
- ▶ Most of my recommended order from research pitch advice holds here
- ▶ Say the topic, give **necessary** background, clearly state the question
- ▶ Probably not time for more than minimal literature review: 1-2 papers
- ▶ No speculation or dithering: what's the data, what's the econometric strategy?

Giving Your Research Presentation (2/2)

- ▶ Show results on **clean** tables: no vertical lines, 3 digits, include (s.e.), highlight?
- ▶ Figures: 1 per slide (2 rarely), label axes and plots, big font, aesthetics matter
- ▶ Interpretation: don't just give numeric results, unpack into sentence about reality
- ▶ Economic “register” loop: reality $\rightarrow$ theory $\rightarrow$ econometrics $\rightarrow$ reality
- ▶ Quick conclusion to wrap up; **maybe** time to say what you'd do next
- ▶ It's polite to introduce yourselves and mention mentors (if they helped)

Student Research Questions and Discussion

Let's talk!